CS201 Midterm 1 Part B

11 July 2024

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***Abstract*— Write only ONE function per question. Do not check the input, assume that the input is correct. Unless noted, (i) the input data structure is not empty, (ii) time complexity of the algorithm does not matter, (iii) you can not use extra data structures including arrays, (iv) you can use any function given in the data structure, (v) you can not modify the data structure contents.**

I. Question (LinkedList) (16 pts)

Write the method

```
LinkedList getIndexed(LinkedList list)
```

which returns the elements at positions list[1], list[2], list[3], etc. list[1] is the first element in the list, list[2] is the second element in the list etc. You are not allowed to use any linked list methods. You are only allowed to use attributes, constructors, getters and setters. Assume that list is sorted. Your algorithm should run in $\mathcal{O}(N)$ time. Your linked list should contain new nodes, not the same nodes in the original linked list.
Contents of the original list
3 1 7 5 11 14 2 8 16
Contents of the list
1 4 6 9
Contents of the results
3 5 14 16

II. Question (DoublyLinkedList) (22 pts)

Write the method

```
void removeKthBeforeLast(int K)
```

which removes the K'th element from the end of the double linked list. If K = 1, last element will be deleted. If K = N, first element will be deleted. First count the number of elements in the list, i.e. N, then handle special cases K = 1, and K = N, then do the rest. You are not allowed to use any doubly linked list methods. You are allowed to use attributes, constructors, getters and setters.

III. Question (Stack) (12 pts)

Write a static method using an external stack (only one external stack is allowed) that determines if an integer array is balanced or not. A number k less than 10 is balanced with the number $10+k$. For example, the array 2, 3, 13, 12, 4, 14 is balanced, whereas 5, 15, 4, 3, 14, 13 not. You are not allowed to use any stack attributes such as N, top, array etc. Write the method in array implementation.

```
boolean isBalanced(int[] a)
```

IV. Question (Stack) (10 pts)

Write the method

```
void addToStack(Stack s, int p, int q)
```

which adds all elements indexed from p to q of the stack s to the top of the original stack. Your stack should contain new nodes, not the nodes in the stack s. The element at the top of the stack has index 1. You are not allowed to use any stack methods and any external stacks. You are allowed to use attributes, constructors, getters and setters. Write the method for linked list implementation.
Contents of the original stack
1 4 5 2
Contents of the stack s
3 1 7 5 11 14 6 8 16
After addToStack(s, 3, 6)
1 4 5 2 6 14 11 5